Lecture 16: Eigenspaces and Diagonalization

MATH203 Applied Linear Algebra

Updated May 6, 2026

What Does P_i Project Onto?

Standing Assumption

Throughout this lecture:

$$\det(tI - A) = (t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_m), \qquad \lambda_i \text{ all distinct}$$

Same assumption as Lecture 15.

We have the spectral decomposition:

$$A = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_m P_m$$

where $P_1, \ldots, P_m$ are compatible projections that partition I.

Today's Question

Each P_i is a projection: $P_i^2 = P_i$.

Every projection projects **onto** some subspace — its column space $\mathrm{Col}(P_i)$.

$$\mathbf{v} = P_1\mathbf{v} + P_2\mathbf{v} + \cdots + P_m\mathbf{v}$$

$$P_i\mathbf{v} \in \mathrm{Col}(P_i)$$

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$$P_i\mathbf{v} \in \mathrm{Col}(P_i)$$

What is $\mathrm{Col}(P_i)$?

What does A do to vectors in $\mathrm{Col}(P_i)$?

A New Example

$$\underbrace{A = \begin{pmatrix} -1 & 2 \\ -4 & 5 \end{pmatrix}}_{\text{our running example}}, \qquad \det(tI - A) = (t-1)(t-3)$$

Eigenvalues: $\lambda_1 = 1$, $\lambda_2 = 3$.

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Eigenvalues: $\lambda_1 = 1$, $\lambda_2 = 3$.

Spectral projections (Lagrange formula from L15):

$$P_1 = \frac{A - 3I}{1 - 3} = \begin{pmatrix} 2 & -1 \\ 2 & -1 \end{pmatrix}, \qquad P_3 = \frac{A - I}{3 - 1} = \begin{pmatrix} -1 & 1 \\ -2 & 2 \end{pmatrix}$$

Value tables (from L15):

	$t = 1$	$t = 3$
A	1	3
P_1	**1**	0
P_3	0	**1**

$$A = \underbrace{1}_{\lambda_1} \cdot P_1 + \underbrace{3}_{\lambda_2} \cdot P_3$$

What Does P_i Do to a Vector?

Pick $\mathbf{v} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$. Apply each projection:

$$P_1\mathbf{v} = \begin{pmatrix} 2 & -1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$P_3\mathbf{v} = \begin{pmatrix} -1 & 1 \\ -2 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

What Does P_i Do to a Vector?

Pick $\mathbf{v} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$. Apply each projection:

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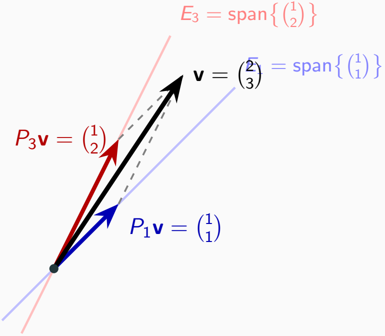
$$P_3\mathbf{v} = \begin{pmatrix} -1 & 1 \\ -2 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Check: $\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \mathbf{v}$ ✓

Every Vector Splits Into Eigenspace Pieces

$$\mathbf{v} = P_1\mathbf{v} + P_3\mathbf{v} = \begin{pmatrix}1\\1\end{pmatrix} + \begin{pmatrix}1\\2\end{pmatrix} = \begin{pmatrix}2\\3\end{pmatrix} \checkmark$$



$P_1 + P_3 = I$: every vector decomposes into eigenspace pieces (parallelogram law).

What Does A Do to Each Piece?

Apply A to the first piece:

$$A \cdot P_1\mathbf{v} = \begin{pmatrix} -1 & 2 \\ -4 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \mathbf{1} \cdot P_1\mathbf{v}$$

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Apply A to the second piece:

$$A \cdot P_3\mathbf{v} = \begin{pmatrix} -1 & 2 \\ -4 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix} = \mathbf{3} \cdot P_3\mathbf{v}$$

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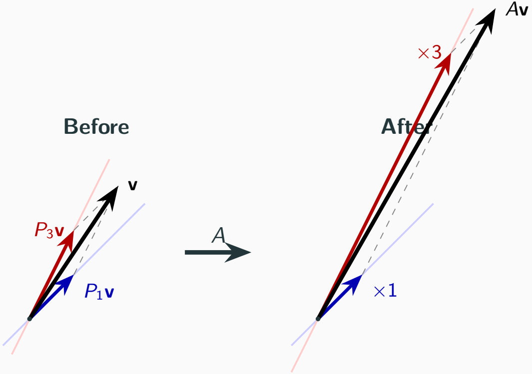
A **scales each piece by its eigenvalue**:
$\lambda_1 = 1$ and $\lambda_2 = 3$.

A Scales Each Piece by Its Eigenvalue

$$A\mathbf{v} \;=\; A\big(P_1\mathbf{v} + P_3\mathbf{v}\big) \;=\; \underbrace{\mathbf{1}}_{\lambda_1}\cdot P_1\mathbf{v} \;+\; \underbrace{\mathbf{3}}_{\lambda_2}\cdot P_3\mathbf{v}$$



A doesn't rotate or mix the pieces — it **only scales** each one along its eigenspace.

Why? — Value Table Proof

Why does A scale the P_1-piece by λ_1?

Multiply value tables entry-wise:

	$t=1$	$t=3$
A	1	3
$\times\ P_1$	1	0
AP_1	**1**	**0**

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Value table of $\lambda_1 P_1 = 1 \cdot P_1$:

	$t=1$	$t=3$
$1 \cdot P_1$	**1**	**0**

Same value table $\implies$ **same matrix**: $AP_1 = \lambda_1 P_1 = 1 \cdot P_1$.

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Similarly for P_3:

	$t=1$	$t=3$
A	1	3
$\times\ P_3$	0	1
AP_3	**0**	**3**

= value table of $3P_3$ ✓

A Commutes With Each P_i

What about $P_1 A$? **Multiply** value tables in the other order:

	$t = 1$	$t = 3$
P_1	1	0
$\times\ A$	1	3
$P_1 A$	**1**	**0**

Same value table as AP_1! So $P_1 A = AP_1 = \lambda_1 P_1$.

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Same value table as AP_1! So $P_1 A = AP_1 = \lambda_1 P_1$.

Value table multiplication is **entry-wise**: order doesn't matter.

$$AP_i = P_i A = \lambda_i P_i \qquad \text{for each } i$$

A commutes with every spectral projection.

(This fact will matter when we find **left eigenvectors** later.)

From $AP_i = \lambda_i P_i$ to Eigenvectors

Suppose $\mathbf{x} \in \mathrm{Col}(P_i)$.

Since P_i is a projection, it **fixes** its own column space: $P_i\mathbf{x} = \mathbf{x}$.

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Suppose $\mathbf{x} \in \mathrm{Col}(P_i)$.

Since P_i is a projection, it **fixes** its own column space: $P_i\mathbf{x} = \mathbf{x}$.

Then:

$$A\mathbf{x} = A\underbrace{P_i\mathbf{x}}_{=\mathbf{x}} = \underbrace{AP_i}_{=\lambda_i P_i}\mathbf{x} = \lambda_i \underbrace{P_i\mathbf{x}}_{=\mathbf{x}} = \lambda_i\mathbf{x}$$

Every vector in $\mathrm{Col}(P_i)$ is scaled by λ_i.

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Every vector in $\mathrm{Col}(P_i)$ is scaled by λ_i.

$\mathrm{Col}(P_i)$ = the subspace where A acts as

scalar multiplication by λ_i.

The Eigenspace

Definition (Eigenspace)

The **eigenspace** of A for eigenvalue λ_i is

$$E_{\lambda_i} \;=\; \left\{\, \mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \lambda_i \mathbf{x} \,\right\}$$

The set of all vectors that A simply **scales** by λ_i.

Proposition

$E_{\lambda_i} = \mathrm{Col}(P_i)$ (the column space of the spectral projection).

Proof on next slide.

Proof: $E_{\lambda_i} = \mathrm{Col}(P_i)$

Proof ($\supseteq$): If $\mathbf{x} \in \mathrm{Col}(P_i)$, then $P_i\mathbf{x} = \mathbf{x}$ (projection fixes its column space), so

$$A\mathbf{x} = AP_i\mathbf{x} = \lambda_i P_i\mathbf{x} = \lambda_i\mathbf{x} \quad \Longrightarrow \quad \mathbf{x} \in E_{\lambda_i}$$

Proof: $E_{\lambda_i} = \mathrm{Col}(P_i)$

Proof ($\supseteq$): If $\mathbf{x} \in \mathrm{Col}(P_i)$, then $P_i\mathbf{x} = \mathbf{x}$ (projection fixes its column space), so

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Proof ($\subseteq$): P_i is a **polynomial** of A (Lagrange interpolation formula from L15).

For any polynomial f of A: if $A\mathbf{x} = \lambda_i\mathbf{x}$, then $f(A)\mathbf{x} = f(\lambda_i)\mathbf{x}$.

Since P_i is the Lagrange polynomial that evaluates to 1 at λ_i:

$$P_i\mathbf{x} = 1 \cdot \mathbf{x} = \mathbf{x} \quad \Longrightarrow \quad \mathbf{x} \in \mathrm{Col}(P_i) \quad \square$$

Check on Our Example

$$E_1 = \mathrm{Col}(P_1) = \mathrm{span}\left\{\begin{pmatrix}1\\1\end{pmatrix}\right\}. \quad A\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}1\\1\end{pmatrix} = \mathbf{1}\begin{pmatrix}1\\1\end{pmatrix} \ \checkmark$$

$$E_3 = \mathrm{Col}(P_3) = \mathrm{span}\left\{\begin{pmatrix}1\\2\end{pmatrix}\right\}. \quad A\begin{pmatrix}1\\2\end{pmatrix} = \begin{pmatrix}3\\6\end{pmatrix} = \mathbf{3}\begin{pmatrix}1\\2\end{pmatrix} \ \checkmark$$

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The eigenspace E_{λ_i} is **exactly** the column space of P_i.

The spectral projections **project onto** the eigenspaces.

P_i projects onto E_{λ_i}:

the subspace where A acts as

scalar multiplication by λ_i.

Compatible Projections Revisited

Recall: Where Did P_1, P_3 Come From?

In Lecture 15, we plugged A into the **Lagrange interpolation polynomials**:

$$P_1 = \frac{A - 3I}{1 - 3}, \qquad P_3 = \frac{A - I}{3 - 1}$$

This gave the **spectral decomposition**: $A = 1 \cdot P_1 + 3 \cdot P_3$.

Their **value tables** (from Lecture 15):

	$t = 1$	$t = 3$
P_1	1	0
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We **claimed** P_1, P_3 are compatible projections that partition I.

Let's verify this using value tables — the tool we trust from Lecture 15.

Are P_1, P_3 Projections?

Check with value tables — **square each entry**:

	$t=1$	$t=3$
P_1	1	0
P_1^2	$1^2 = \mathbf{1}$	$0^2 = \mathbf{0}$

	$t=1$	$t=3$
P_3	0	1
P_3^2	$0^2 = \mathbf{0}$	$1^2 = \mathbf{1}$

Same value tables $\implies$ $P_1^2 = P_1$ and $P_3^2 = P_3$. Both are **projections**. ✓

Compatible? Partition of I?

Compatible? Multiply value tables entry-wise:

	$t = 1$	$t = 3$
P_1	1	0
$\times\ P_3$	0	1
$P_1 P_3$	**0**	**0**

Value table all zeros $\implies$ $P_1 P_3 = 0$. **Compatible!** ✓

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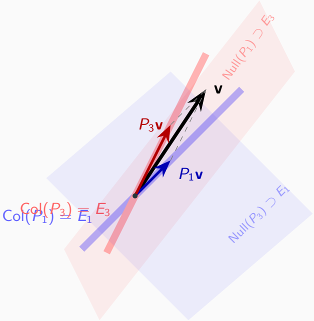
Partition of I? Add value tables:

	$t = 1$	$t = 3$
P_1	1	0
$+\ P_3$	0	1
$P_1 + P_3$	**1**	**1**

Value table $=$ value table of I. $P_1 + P_3 = I$. ✓

Connection to Lecture 8: Compatible Projections

From Lecture 8: $P_1 P_3 = 0$ means $\mathrm{Col}(P_3) \subset \mathrm{Null}(P_1)$ and $\mathrm{Col}(P_1) \subset \mathrm{Null}(P_3)$.



Each eigenspace lives inside the null space of the **other** projection.

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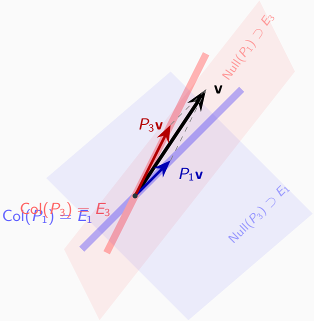

Each eigenspace lives inside the null space of the **other** projection.

The Lagrange formula from L15 **automatically**

produces compatible projections.

No extra verification needed — it's built into the construction.

Eigenvectors With Different Eigenvalues Are Independent

From Lecture 9, Proposition (Independent Decompositions):

For a compatible family $\{P_1, \ldots, P_m\}$, the nonzero vectors among $\{P_1\mathbf{v}, \ldots, P_m\mathbf{v}\}$ are **linearly independent**.

Proof: multiply both sides of $\sum c_i P_i \mathbf{v} = 0$ by P_j. Since $P_j P_i = 0$ for $j \neq i$, only the j-th term survives: $c_j P_j \mathbf{v} = 0$.

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Since the spectral projections $P_1, \ldots, P_m$ are compatible:

Corollary

Let $\mathbf{v}_1, \ldots, \mathbf{v}_m$ be eigenvectors of A with **distinct** eigenvalues $\lambda_1, \ldots, \lambda_m$. Then $\mathbf{v}_1, \ldots, \mathbf{v}_m$ are **linearly independent**.

Each $\mathbf{v}_i \in E_{\lambda_i} = \mathrm{Col}(P_i)$, so $P_i\mathbf{v}_i = \mathbf{v}_i \neq 0$ and $P_j\mathbf{v}_i = 0$ for $j \neq i$.

The Lecture 9 proposition applies directly. $\square$

Unique Decomposition (Lecture 9 Callback)

From Lecture 9: compatible projections that partition I give a **unique decomposition** of every vector.

Since $P_1 + P_3 = I$, every $\mathbf{v} \in \mathbb{R}^n$ decomposes as:

$$\mathbf{v} = \underbrace{P_1\mathbf{v}}_{\in E_1} + \underbrace{P_3\mathbf{v}}_{\in E_3}$$

and by the corollary above, this decomposition is **unique**.

The eigenspaces $E_{\lambda_1}, \ldots, E_{\lambda_m}$ together **span** all of $\mathbb{R}^n$.

Every vector decomposes **uniquely** into eigenvector pieces.

(This is the geometric meaning of $P_1 + P_2 + \cdots + P_m = I$.)

Cross-Filling the Spectral Projections

A New Example — Repeated Eigenvalue

$$\underbrace{B = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 5 \end{pmatrix}}_{\text{eigenvalues } 2,2,5}, \qquad \det(tI - B) = (t-2)^2(t-5)$$

Two **distinct** eigenvalues: $\lambda_1 = 2$, $\lambda_2 = 5$.

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Two **distinct** eigenvalues: $\lambda_1 = 2$, $\lambda_2 = 5$.

Spectral projections (Lagrange formula):

$$P_5 = \frac{B - 2I}{5 - 2} = \frac{1}{3}\begin{pmatrix} 0 & 0 & 3 \\ 0 & 0 & 3 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$P_2 = I - P_5 = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

	$t = 2$	$t = 5$
B	2	5
P_2	**1**	0
P_5	0	**1**

P_2 Has Rank 2 — A Plane

$$P_2 = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \qquad \text{rank} = 2$$

$E_2 = \mathrm{Col}(P_2)$ is a **plane** in $\mathbb{R}^3$, not just a line.

$$P_5 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \qquad \text{rank} = 1$$

$E_5 = \mathrm{Col}(P_5)$ is a **line** in $\mathbb{R}^3$.

P_2 Has Rank 2 — A Plane

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$E_5 = \mathrm{Col}(P_5)$ is a **line** in $\mathbb{R}^3$.

To find eigenvectors, we **cross-fill** each projection (Lecture 8).

Cross-Fill P_2 — Step 1

$$P_2 = \begin{pmatrix} \mathbf{1} & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

Pivot at position $(1,1)$: $\mathbf{1}$. Column through pivot: $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$. Row through pivot: $\begin{pmatrix} 1 & 0 & -1 \end{pmatrix}$.

Rank-1 piece (outer product of column $\times$ row):

$$R_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \end{pmatrix} = \begin{pmatrix} \mathbf{1} & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Remainder: $P_2 - R_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$

Cross-Fill P_2 — Step 2 and Assembly

Remainder: $\begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{1} & -1 \\ 0 & 0 & 0 \end{pmatrix}$. Pivot at $(2,2)$: $\mathbf{1}$.

$$R_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{1} & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

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Matrix sum (colored cross-filling decomposition):

$$P_2 = \underbrace{\begin{pmatrix} \mathbf{1} & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{R_1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{1} & -1 \\ 0 & 0 & 0 \end{pmatrix}}_{R_2}$$

Two-factor form (stack columns and rows):

$$P_2 = \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}}_{U_2} \underbrace{\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix}}_{V_2}, \qquad V_2 U_2 = I_2 \checkmark$$

Cross-Fill P_5 — One Step

$$P_5 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & \mathbf{1} \end{pmatrix}$$

Pivot at $(3, 3)$: **1**.

$$P_5 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}_{U_5} \underbrace{\begin{pmatrix} 0 & 0 & 1 \end{pmatrix}}_{V_5}$$

Check: $V_5 U_5 = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 1 \ \checkmark$

Columns = Right Eigenvectors, Rows = Left Eigenvectors

Right eigenvectors (columns of U_i): verify $B\mathbf{u} = \lambda\mathbf{u}$:

$$B\begin{pmatrix}1\\0\\0\end{pmatrix} = \mathbf{2}\begin{pmatrix}1\\0\\0\end{pmatrix} \checkmark \qquad B\begin{pmatrix}0\\1\\0\end{pmatrix} = \mathbf{2}\begin{pmatrix}0\\1\\0\end{pmatrix} \checkmark \qquad B\begin{pmatrix}1\\1\\1\end{pmatrix} = \mathbf{5}\begin{pmatrix}1\\1\\1\end{pmatrix} \checkmark$$

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Left eigenvectors (rows of V_i): verify $\mathbf{v}^T B = \lambda\mathbf{v}^T$:

$$\begin{pmatrix}1 & 0 & -1\end{pmatrix} B = \mathbf{2}\begin{pmatrix}1 & 0 & -1\end{pmatrix}\ \checkmark \quad \begin{pmatrix}0 & 1 & -1\end{pmatrix} B = \mathbf{2}\begin{pmatrix}0 & 1 & -1\end{pmatrix}\ \checkmark$$
$$\begin{pmatrix}0 & 0 & 1\end{pmatrix} B = \mathbf{5}\begin{pmatrix}0 & 0 & 1\end{pmatrix}\ \checkmark$$

Columns = Right Eigenvectors, Rows = Left Eigenvectors

Right eigenvectors (columns of U_i): verify $B\mathbf{u} = \lambda\mathbf{u}$:

$$B\begin{pmatrix}1\\0\\0\end{pmatrix} = \mathbf{2}\begin{pmatrix}1\\0\\0\end{pmatrix} \checkmark \qquad B\begin{pmatrix}0\\1\\0\end{pmatrix} = \mathbf{2}\begin{pmatrix}0\\1\\0\end{pmatrix} \checkmark \qquad B\begin{pmatrix}1\\1\\1\end{pmatrix} = \mathbf{5}\begin{pmatrix}1\\1\\1\end{pmatrix} \checkmark$$

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Cross-eigenspace: $V_2U_5 = 0$, $V_5U_2 = 0$ (from $P_2P_5 = 0$, as in §2).

Recall: Cross-Filling Produces Rank-1 Projections (L08, L09)

From Lecture 8: cross-filling a projection gives rank-1 pieces R_i that are **themselves projections**.

From Lecture 9 (Corollary):

> Let P be a projection. Cross-fill $P = R_1 + \cdots + R_r$ into rank-1 pieces.
>
> Then $\{R_1, \ldots, R_r\}$ is a **compatible family of rank-1 projections**.
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> (Proof: $\sum \text{rank}(R_i) = r = \text{rank}(P)$, apply Projection Decomposition Theorem.)

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For our P_2 (rank 2):

$$P_2 = \underbrace{\begin{pmatrix} \mathbf{1} & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{R_1 \text{ (rank 1, projection)}} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{1} & -1 \\ 0 & 0 & 0 \end{pmatrix}}_{R_2 \text{ (rank 1, projection)}}$$

R_1, R_2 are compatible: $R_1 R_2 = 0$, $R_1^2 = R_1$, $R_2^2 = R_2$. And P_5 is already rank 1.

The Final Spectral Decomposition

Start from: $B = 2P_2 + 5P_5$.

Replace $P_2 = R_1 + R_2$ (its rank-1 pieces). Now **every piece is rank 1**:

$$B \;=\; 2 \cdot R_1 \;+\; 2 \cdot R_2 \;+\; 5 \cdot P_5$$

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Write out each rank-1 piece as column $\times$ row:

$$B \;=\; 2 \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \end{pmatrix} + 2 \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & -1 \end{pmatrix} + 5 \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}$$

Three colored rank-1 pieces — one per eigenvector.

From Lecture 8: a sum of rank-1 matrices becomes a **product** $\longrightarrow$

From Sum to Product (The Cross-Filling Spirit)

Stack columns into P, scalars into D, rows into Q:

$$B = 2 \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \end{pmatrix} + 2 \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & -1 \end{pmatrix} + 5 \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}$$

$$= \underbrace{\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}}_{P \text{ (columns)}} \underbrace{\begin{pmatrix} 2 & & \\ & 2 & \\ & & 5 \end{pmatrix}}_{D \text{ (scalars)}} \underbrace{\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}}_{Q \text{ (rows)}}$$

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Each column of P pairs with the matching row of Q and scalar in D.

The General Formula

$$B = \lambda_1 \mathbf{u}_1 \mathbf{v}_1^T + \lambda_2 \mathbf{u}_2 \mathbf{v}_2^T + \lambda_3 \mathbf{u}_3 \mathbf{v}_3^T = P \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{pmatrix} Q$$

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More generally, replace λ_i by $g(\lambda_i)$:

$$g(B) = g(\lambda_1)\,\mathbf{u}_1\mathbf{v}_1^T + g(\lambda_2)\,\mathbf{u}_2\mathbf{v}_2^T + g(\lambda_3)\,\mathbf{u}_3\mathbf{v}_3^T = P\begin{pmatrix}g(\lambda_1) & & \\ & g(\lambda_2) & \\ & & g(\lambda_3)\end{pmatrix}Q$$

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Same columns, same rows — only the **scalars on the diagonal change**.

$$g(\lambda) = \lambda^n: \qquad B^n = P \begin{pmatrix} 2^n & & \\ & 2^n & \\ & & 5^n \end{pmatrix} P^{-1}$$

Why $Q = P^{-1}$ (Set $g \equiv 1$)

Apply the general formula with $g(\lambda) = 1$ for all λ:

$$g(B) = g(\lambda_1)\,\mathbf{u}_1\mathbf{v}_1^T + g(\lambda_2)\,\mathbf{u}_2\mathbf{v}_2^T + g(\lambda_3)\,\mathbf{u}_3\mathbf{v}_3^T = P \begin{pmatrix} g(\lambda_1) & & \\ & g(\lambda_2) & \\ & & g(\lambda_3) \end{pmatrix} Q$$

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Left side: $R_1 + R_2 + P_5 = P_2 + P_5 = I$ (value table from §2). ✓

$$\boxed{PQ = I, \qquad \text{so} \quad Q = P^{-1}}$$

Diagonalization: $B = PDP^{-1}$

$$\underbrace{\begin{pmatrix} 2 & 0 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 5 \end{pmatrix}}_{B} = \underbrace{\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}}_{P} \underbrace{\begin{pmatrix} 2 & & \\ & 2 & \\ & & 5 \end{pmatrix}}_{D} \underbrace{\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}}_{P^{-1}}$$

Verify: $PD = \begin{pmatrix} 2 & 0 & 5 \\ 0 & 2 & 5 \\ 0 & 0 & 5 \end{pmatrix}, \quad PDP^{-1} = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 5 \end{pmatrix} = B \ \checkmark$

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Columns of P = right eigenvectors. Rows of P^{-1} = left eigenvectors.

Diagonal of D = eigenvalues (with repetition).

The Cancellation Trick

Why does $g(B) = P\,g(D)\,P^{-1}$?

$$B^2 = (\overbrace{PDP^{-1}}^{B})(\overbrace{PDP^{-1}}^{B}) = PD\,\underbrace{\cancel{P^{-1}P}}_{=I}\,DP^{-1} = PD^2P^{-1}$$

The $P^{-1}P$ in the middle **cancels**!

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Powers:

$$B^n = PD^nP^{-1}, \qquad D^n = \begin{pmatrix} 2^n & & \\ & 2^n & \\ & & 5^n \end{pmatrix}$$

Diagonal matrices are trivial to exponentiate.

Connection to Value Tables

Two notations for the same machine:

Spectral formula (L15): $g(B) = g(2)\, P_2 + g(5)\, P_5$

Diagonalization: $g(B) = P\, g(D)\, P^{-1}$

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Spectral formula (L15): $g(B) = g(2)\, P_2 + g(5)\, P_5$

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	$t = 2$	$t = 5$
$g(B)$	$g(2)$	$g(5)$

The value table computes $g(D)$. The spectral projections are hidden inside P and P^{-1}.

Same machine, two notations.

Eigenspace Geometry: The Dominant Eigenvalue

A^n via Value Tables

Back to our 2×2 example: $A = \begin{pmatrix} -1 & 2 \\ -4 & 5 \end{pmatrix}$, eigenvalues $1, 3$.

	$t = 1$	$t = 3$
A	1	3
A^2	1	9
A^3	1	27
A^n	1	3^n

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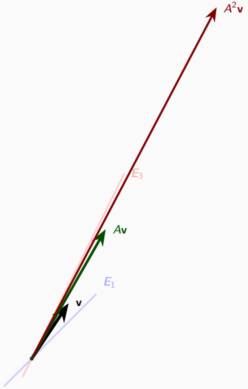
For $\mathbf{v} = \binom{2}{3}$: $\quad A^n\mathbf{v} = \underbrace{\underbrace{1^n}_{=1}\begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{P_1\mathbf{v}} + \underbrace{\underbrace{3^n}_{\to\infty}\begin{pmatrix} 1 \\ 2 \end{pmatrix}}_{P_3\mathbf{v}}$

As $n \to \infty$: the E_3 piece grows like 3^n, the E_1 piece stays $= 1$.

The dominant eigenvalue wins: $A^n\mathbf{v}$ aligns with $E_3 = \operatorname{span}\left\{\binom{1}{2}\right\}$.

Visualizing A^n



Each step: the E_3 piece stretches $\times 3$, the E_1 piece is unchanged.

Eigenvalues tell you **how much** each direction stretches.

Eigenspaces tell you **which** directions.

Together, they completely describe what A does.

Our Restriction: The Minimal Polynomial

A Repeated Root That Worked

B from §3 had $\det(tI - B) = (t-2)^{2}(t-5)$ — a **repeated** root!

Yet the spectral machinery worked perfectly:

- Two distinct projections P_2, P_5 ✓
- Compatible, partition of I ✓
- Cross-filling, eigenvectors ✓
- $B = PDP^{-1}$ ✓

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- Two distinct projections P_2, P_5 ✓
- Compatible, partition of I ✓
- Cross-filling, eigenvectors ✓
- $B = PDP^{-1}$ ✓

Why? The characteristic polynomial can repeat. What matters is a different polynomial.

The Minimal Polynomial

Definition (Minimal Polynomial)

The **minimal polynomial** $m(t)$ is the lowest-degree monic polynomial such that $m(A) = 0$.

It divides every annihilating polynomial (including $\det(tI - A)$).

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Check on B:

$$(B - 2I)(B - 5I) = \begin{pmatrix} 0 & 0 & 3 \\ 0 & 0 & 3 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} -3 & 0 & 3 \\ 0 & -3 & 3 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

So $m(t) = (t - 2)(t - 5)$ — **distinct** linear factors. ✓

Another Fine Example: $A = 3I$

$$A = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}, \qquad \det(tI - A) = (t-3)^2 \quad \text{(repeated!)}$$

But $A - 3I = 0$ already, so $m(t) = t - 3$ (degree 1, one distinct factor).

Spectral decomposition: $A = 3 \cdot I$. Just one projection $P_1 = I$.

The whole space $\mathbb{R}^2$ is one eigenspace. Fine!

Beyond Our Assumption: The Shear

$$J = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}, \qquad \det(tI - J) = (t-2)^2$$

$$J - 2I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq 0. \qquad (J-2I)^2 = 0.$$

So $m(t) = (t-2)^2$ — **repeated** factor!

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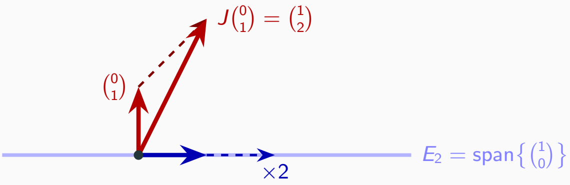

Vectors on E_2 scale by 2. Vectors off E_2 get **sheared** — not just scaled.

This situation needs **new tools** — a topic for a future lecture.

The Real Restriction

The characteristic polynomial CAN have repeated roots.

The **minimal polynomial** is what matters:

Distinct linear factors in $m(t) \implies$ **diagonalizable**.

Repeated factor in $m(t) \implies$ needs new tools (future lecture).

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Repeated factor in $m(t) \implies$ needs new tools (future lecture).

	Char. poly	Min. poly	
$B = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 5 \end{pmatrix}$	$(t-2)^2(t-5)$	$(t-2)(t-5)$	✓
$3I_2$	$(t-3)^2$	$(t-3)$	✓
$J = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$	$(t-2)^2$	$(t-2)^2$	×

Summary

1. $E_{\lambda_i} = \mathrm{Col}(P_i)$: eigenspace = column space of spectral projection.
2. Compatible projections (value tables verify instantly): eigenvectors with different eigenvalues are independent; every vector decomposes uniquely.
3. Cross-fill $P_i = U_i V_i$: columns of U_i = right eigenvectors, rows of V_i = left eigenvectors. $V_i U_j = \delta_{ij} I$.
4. Stack: $A = PDP^{-1}, \quad g(A) = Pg(D)P^{-1}$ reduces every computation to plugging eigenvalues into g.
5. The **minimal polynomial** is the real restriction: distinct linear factors $\implies$ diagonalizable.

Next Lecture

We can compute A^n via eigenvalues. But what about e^A?

$$e^A = \lim_{n \to \infty} \left(I + \frac{A}{n} \right)^n$$

Next: The exponential function, matrix exponential, and differential equations.